WORKFLOW OPERATIONS PERFORMANCE & DELAY ANALYSIS

A Data-Driven Study on Workflow Efficiency, Delay Patterns, and Operational Bottlenecks

AN ANALYTICAL STUDY BY ISHITA MISHRA

Problem Statement

Despite handling a large number of operational tasks, the workflow system experienced consistently high delays in task completion. A significant portion of tasks exceeded their expected completion times, indicating potential inefficiencies in approval processes, workload distribution, and operational workflows.

However, the organization lacked clear visibility into which processes, departments, or task types were contributing most to these delays. Without a data-driven understanding of workflow performance, it was difficult to identify bottlenecks and implement effective improvements

Objectives

- Analyze overall task performance and delay patterns across the workflow system.
- Identify processes and task types with the highest delay rates.
- Evaluate department-level workload and operational costs.
- Examine the relationship between employee workload and task delays.
- Provide data-driven insights and recommendations to improve workflow efficiency.

Tools Used

Used to clean, aggregate, and analyze workflow task data to calculate delay rates, completion times, and identify process bottlenecks.

Used to create the interactive dashboard, visualize KPIs, and present insights through charts and metrics.

SQL Analysis Approach

- SQL was used to clean, transform, and analyze the workflow dataset.
- Queries were written to calculate task delays, process performance, and department costs.
- Aggregations and conditional logic were used to identify patterns and bottlenecks in the workflow.

Key analyses performed using SQL:

- Calculated total tasks, delayed tasks, and delay rate
- Analyzed delay rate by process
- Evaluated department-wise task costs
- Measured average completion time by task type
- Studied employee workload vs delay patterns

Dashboard

Workflow Operations Performance & Delay Analysis

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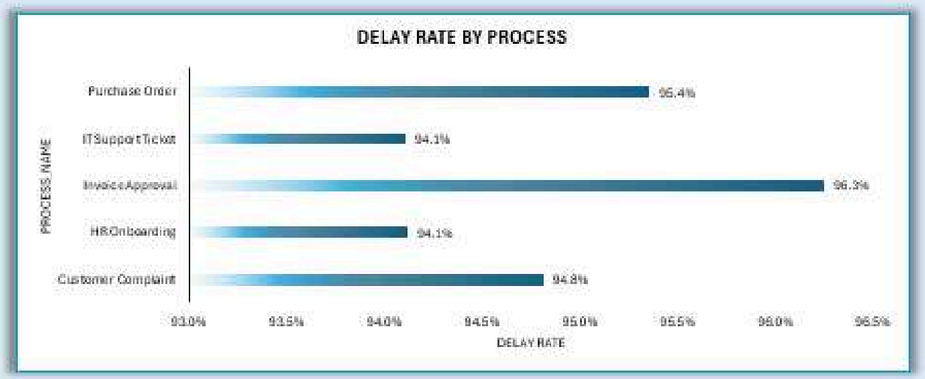
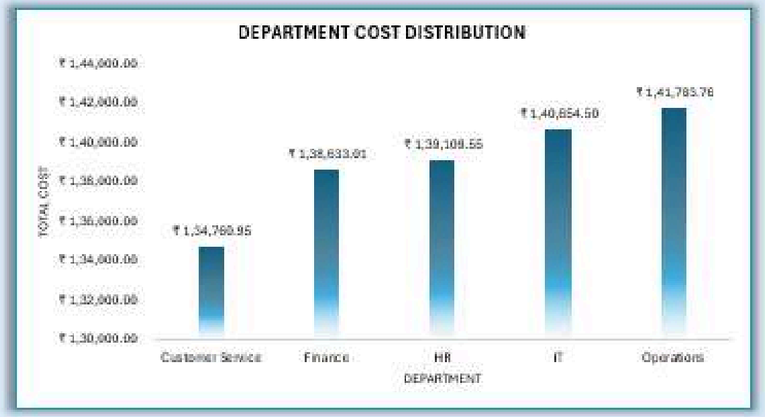
Total Tasks	Total Delays	Delay Rate	Avg Completion Time	Avg Cost Per Task
2500	2373	95%	180.86	277.98

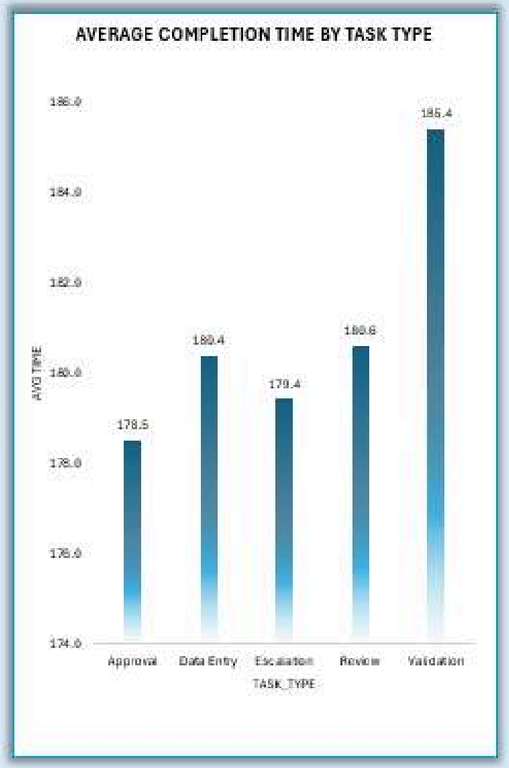
Department: Customer Se... | Finance | HR | IT | Operations

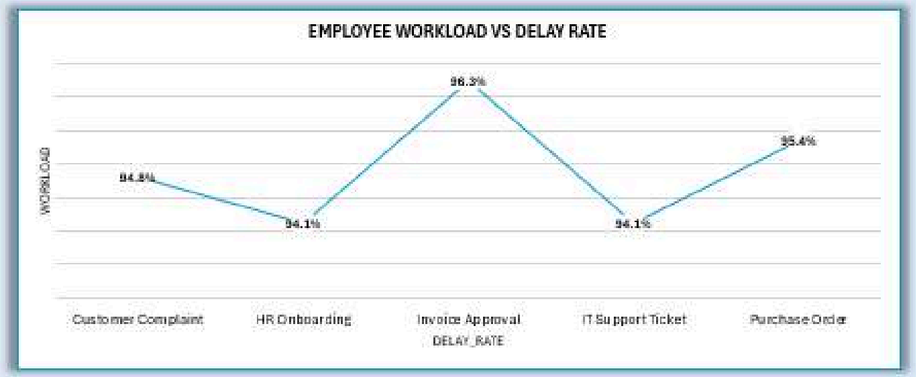
Task_Type: Approval | Data Entry | Escalation | Review | Validation

Priority_Level: Critical | High | Low | Medium







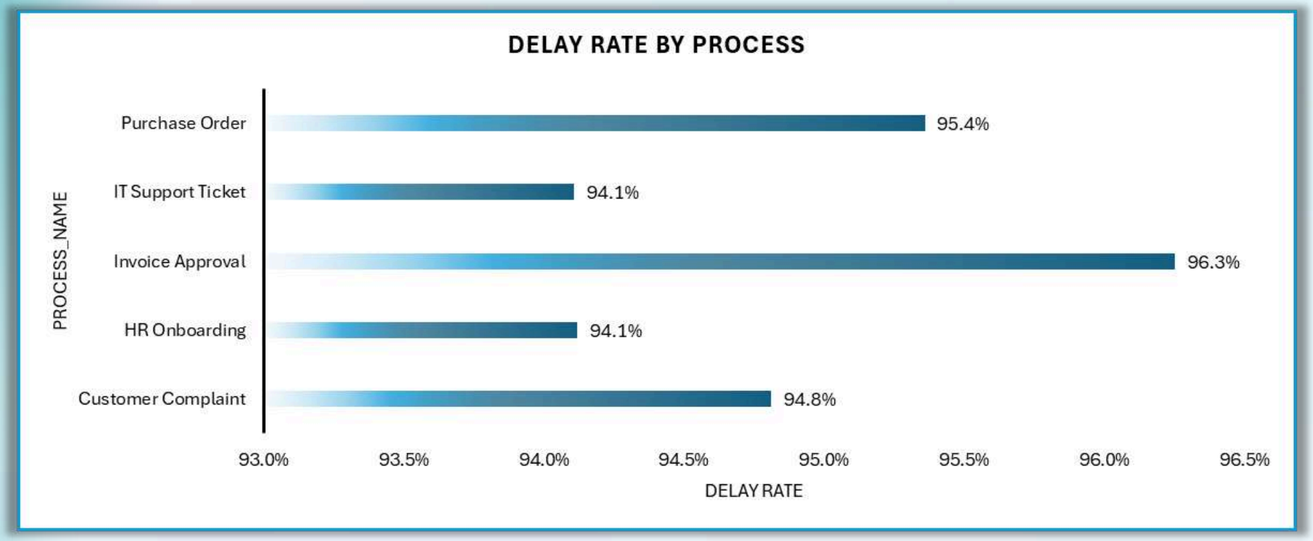


Executive Summary Metrics

Total Tasks	Total Delays	Delay Rate	Avg Completion Time	Avg Cost Per Task
2500	2373	95%	180.86	277.98

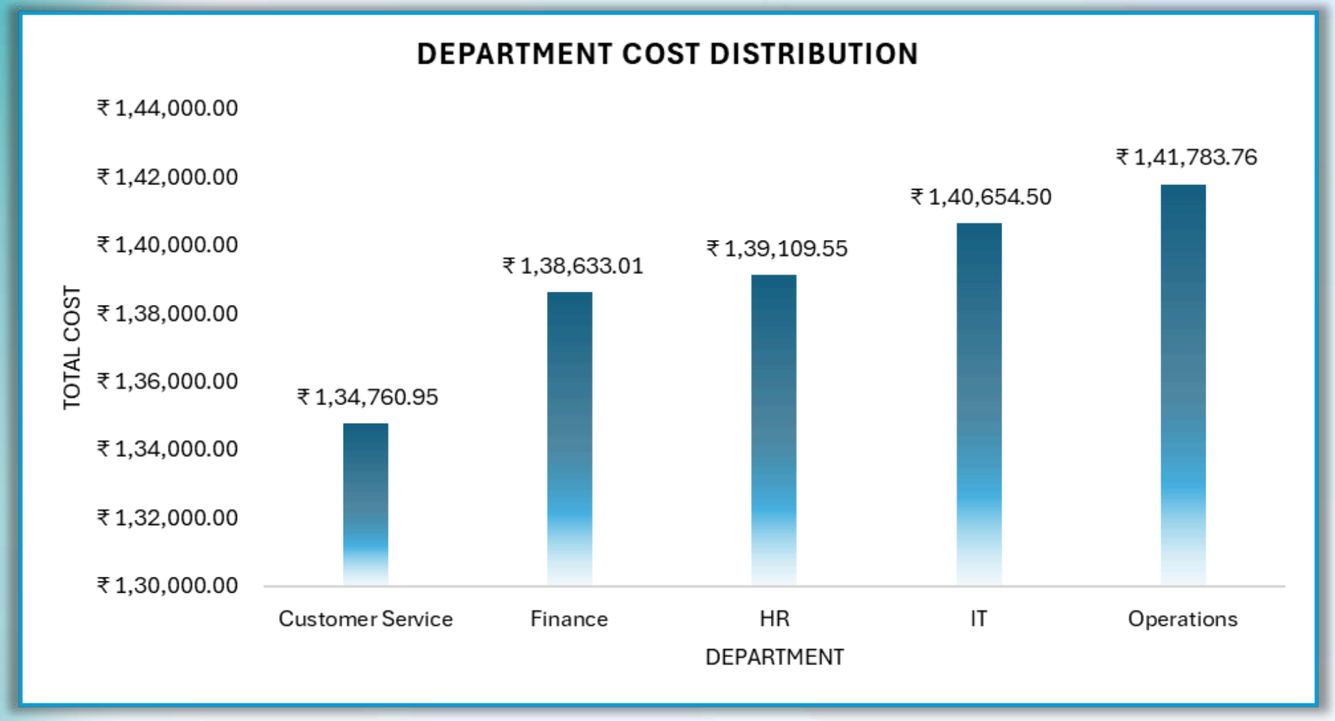
- Total tasks & delays in the system handled a total of 2,500 tasks, of which 2,373 were delayed.
- Success metric of this results in a massive 94.92% delay rate.
- The efficiency average time to complete any task is 180.86 hours, with an average cost of ₹277.98 per task.

Process Performance Analysis



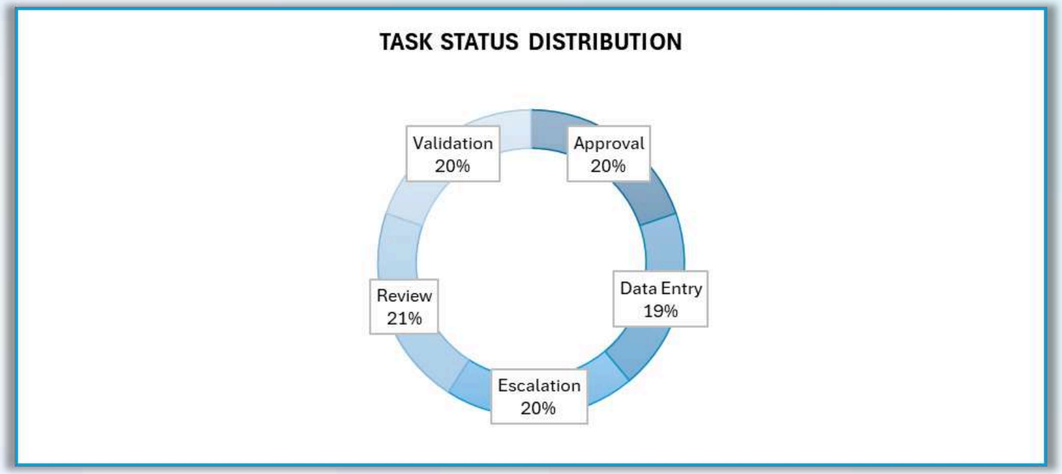
- Invoice Approval has the highest delay rate (~96%)
- Purchase Orders have the highest task volume
- Approval-heavy workflows appear to be major bottlenecks

Department Cost Analysis



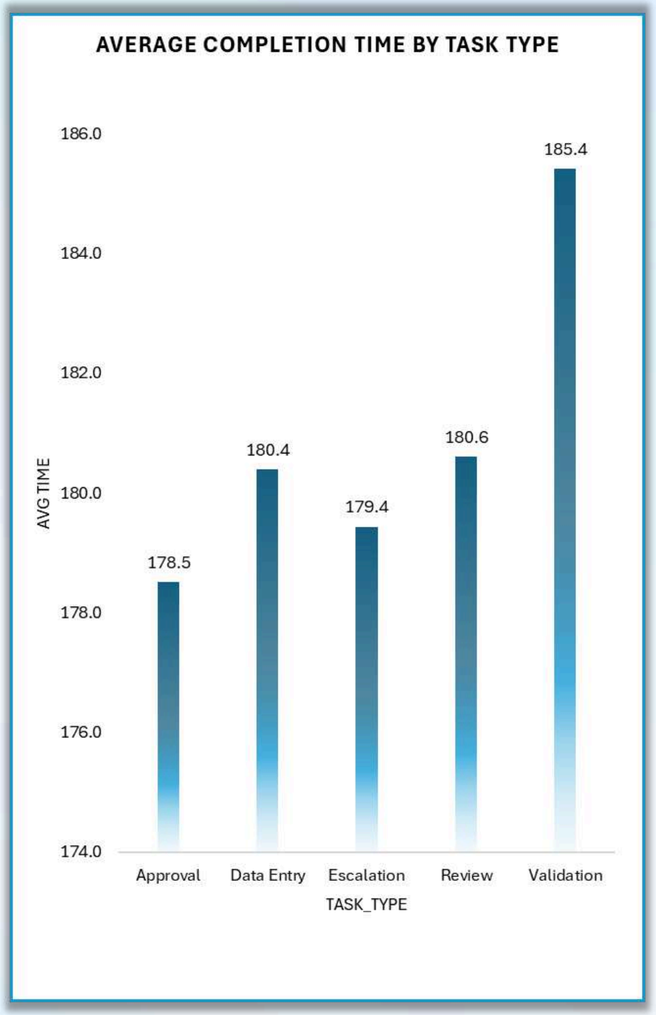
- Operations department has the highest total cost
- IT department also contributes significantly to workflow cost
- Customer Service has the lowest operational cost

Task Type Efficiency



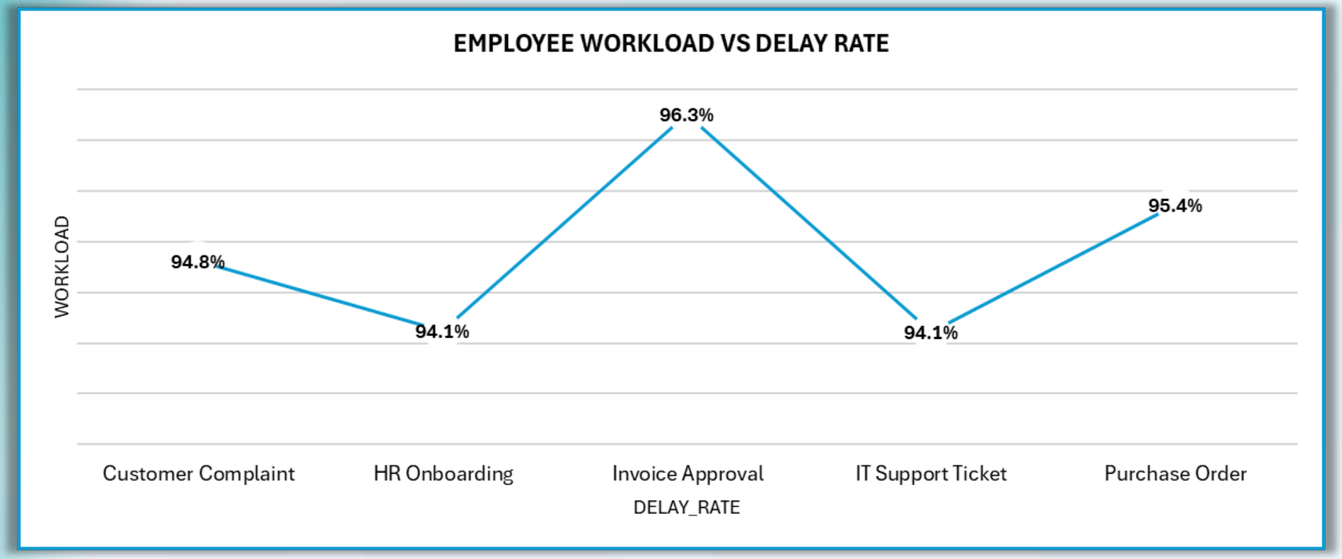
- Tasks are evenly distributed across Approval, Review, Escalation, Validation, and Data Entry.
- Approval and Review make up a large share of tasks.
- Delays likely stem from process inefficiencies, not task volume.

Task Type Efficiency



- Validation tasks take the longest time to complete
- Data Entry tasks are completed faster
- Manual verification steps increase completion time

Workload Impact



- Higher workload levels show higher delay rates
- Workload imbalance may contribute to task delays

Overall Takeaway

- The workflow system shows a very high delay rate across most tasks.
- Approval and validation processes appear to be the main bottlenecks.
- Operations and IT departments contribute the highest operational costs.
- Delays seem to be caused by process inefficiencies rather than task volume.

Recommendations

- Automate approval workflows to reduce delays.
- Simplify validation and review processes to speed up completion.
- Improve workload distribution among employees.
- Use monitoring dashboards to track workflow performance and identify bottlenecks early.

THANK YOU !